

Predicting ATP Singles Match Outcomes with Supervised Learning :
Beating Elo but Not the Market

Antonio Cannito
University of Bari Aldo Moro
Machine Learning Course



A prediction problem with an external referee

Predicting the winner of an ATP singles match is a binary classification task with an unambiguous outcome and public covariates.

An Unusual Benchmark

Bookmakers publish their own probabilities.

Falsifiable by Design

The model is measured against an external competitor, not only against itself.

Value = Information

A model is useful not because it reaches conventional accuracy, but because it carries information the market does not already have.

Related work

Understanding the landscape of sports predictive modeling is crucial for contextualizing novel approaches.
This work intersects with, yet diverges from, established methodologies across three primary families.

- **Point-based hierarchical models** estimate serve and return probabilities, propagating them through the scoring system. While highly interpretable, they necessitate granular point-level data and typically assume independence between points.
- **Rating systems** summarise a player's performance into a single numerical value, updated after each match. The Elo rating system remains a robust single-number predictor, with subsequent refinements often weighting updates based on tournament importance or playing surface to enhance accuracy and relevance.
- **Machine-learning** approaches construct comprehensive feature vectors from extensive match histories, then apply general-purpose classifiers to predict outcomes. These methods leverage complex patterns and relationships within data, often achieving high predictive power but sometimes at the cost of interpretability.
- This work **does not aim** to propose a superior rating system or a novel model architecture. Instead, its core contribution lies in quantifying the disparity between the predictions of a well-constructed supervised model and the actual market outcomes, offering insights into market efficiency and model performance.

From two winner-oriented sources to one symmetric dataset

Raw sources

Match results — 75,010 rows — 51 columns

```
winner_name  loser_name  winner_rank  loser_rank
w_ace  l_ace  surface  score …
```

Bookmaker odds — 66,702 rows — 55 columns

```
Winner  Loser  B365W  B365L  PSW  PSL  Date …
```

No shared identifier. Every column is labelled by the outcome.

Processed dataset

65,281 usable matches — 38 features

```
player_a  player_b  d_elo_blend  d_log_rank
d_winrate_25  surface  y
```

Roles assigned at random, attributes only as differences. Class balance 0.4971.

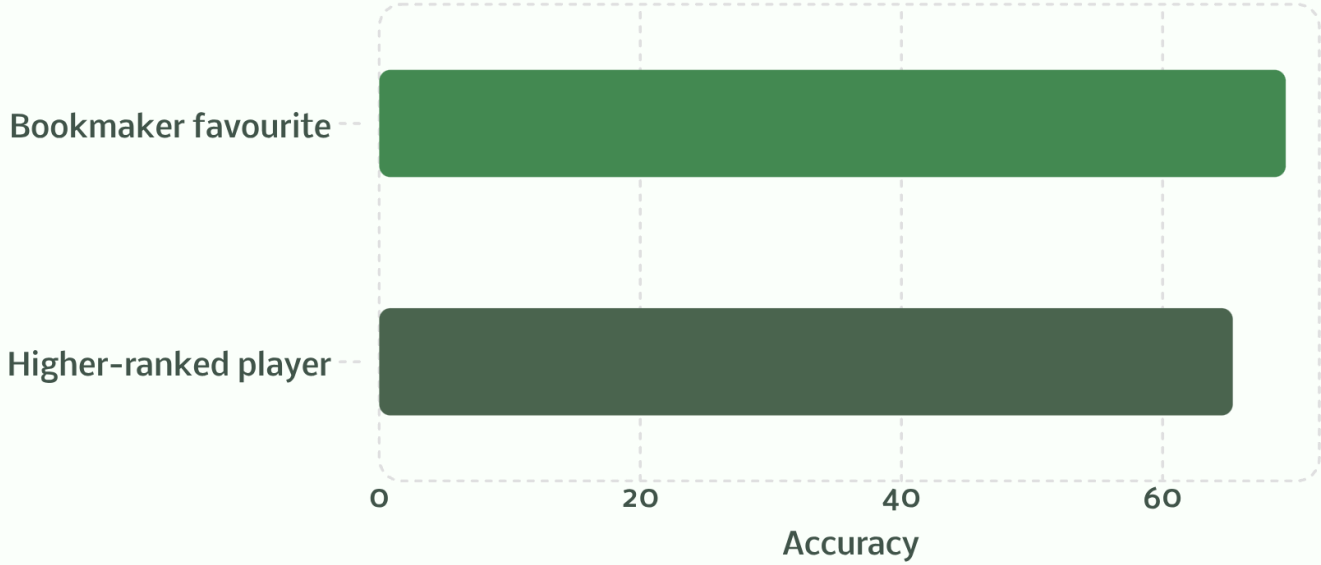
Excluded in sequence: 7,081 Davis Cup ties, 2,436 retirements and walkovers, 212 without a ranking.

Joining the two sources

- No shared identifier and names written differently, so the join key is the player pair plus a 6-day window on the date
- Compound surnames are frequent on tour and blur the forename/surname boundary, so every plausible segmentation of a full name is generated, and the join is accepted if any of them matches
- Coverage with the market: **60,082 of 65,281 matches, 92.0**%

Two baselines

- The higher-ranked player wins 65.37% of matches: no model, no history
- The bookmaker favourite wins 69.43%
- Between them lies the room in which machine learning can operate; measuring how much is recoverable is the question of this work.



Two failure modes that produce no error message

Label ordering

- The winner is always the first column in the raw data
- A feature left oriented as "winner minus loser" predicts the label almost perfectly
- Ruled out by: random A/B slots, features only as differences

Temporal leakage

- Every player attribute is itself computed from past matches
- Computed in one pass over the whole dataset, a 2010 match carries 2015 results
- Ruled out by a single chronological pass, history read before it is updated

Neither raises an exception. Both simply make the numbers better.

Symmetric representation

Eliminating ordering bias in player matchup models through randomized slot assignment and difference-based features.

The Flip Test

Slot A	Slot B	p(A wins)
Player A	Player B	0.63
Player B	Player A	0.34

Same match, swapped order.
These should sum to 1. They sum to 0.97.

After symmetrisation

Slot A	Slot B	p(A wins)
Player A	Player B	0.645
Player B	Player A	0.355

The Trap

In raw data, the winner is always listed first. Any feature built as winner − loser encodes the label outright, leaking the target into the input.

The Fix

Players are assigned to slots A and B at random (fixed seed); y = 1 if A wins.
Class balance: 0.4971.
Attributes enter only as differences d = x_A − x_B, which flip sign when slots are swapped.

The Residue

Not everything flips. The differences do, but the surface, the round, and the model's own baseline are the same in either ordering.
That is what breaks the symmetry.

The Solution

Predict each match in both orderings and average:
$p_{sym} = \frac{1}{2} [\, p(A, B) + 1 - p(B, A) \,]$
The two predictions cancel term by term. The residue becomes exactly zero

Causal features



One pass. The outcome is written only after the features are frozen.

Date alone is not enough

The recorded date is the tournament **START** date, identical for every match of an event. Sorting by date alone could place a final before a first round.

Full ordering:

1. Tournament date
2. Tournament
3. Round
4. Match number

38 Features

Nine blocks, built in one chronological pass.

Block	What it contains
Elo ratings	Global, per surface, and the blend of the two
Rank and points	Log-transformed: the rank–outcome relation saturates beyond about 100 places
Recent form	Win rate over the last 10, 25 and 50 matches, the same restricted to the current surface, and career experience
Load and rest	Matches in the last 30 and 90 days, days since the last match
Injury proxies	Retirement rates, not counts, over 180 and 730 days
Technical stats	Serve and return performance, averaged over the previous 20 matches
Head-to-head	Balance of past meetings plus how many meetings exist
Demographics	Age, height, handedness
Context	Surface, tourney level, round, format, indoor, draw size

30 of the 38 are differences d = xA − xB and flip sign when the slots are swapped. The remaining 8 are shared by both players and do not.

Elo ratings

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

Expected score for A. Ratings move by K × (actual − expected).

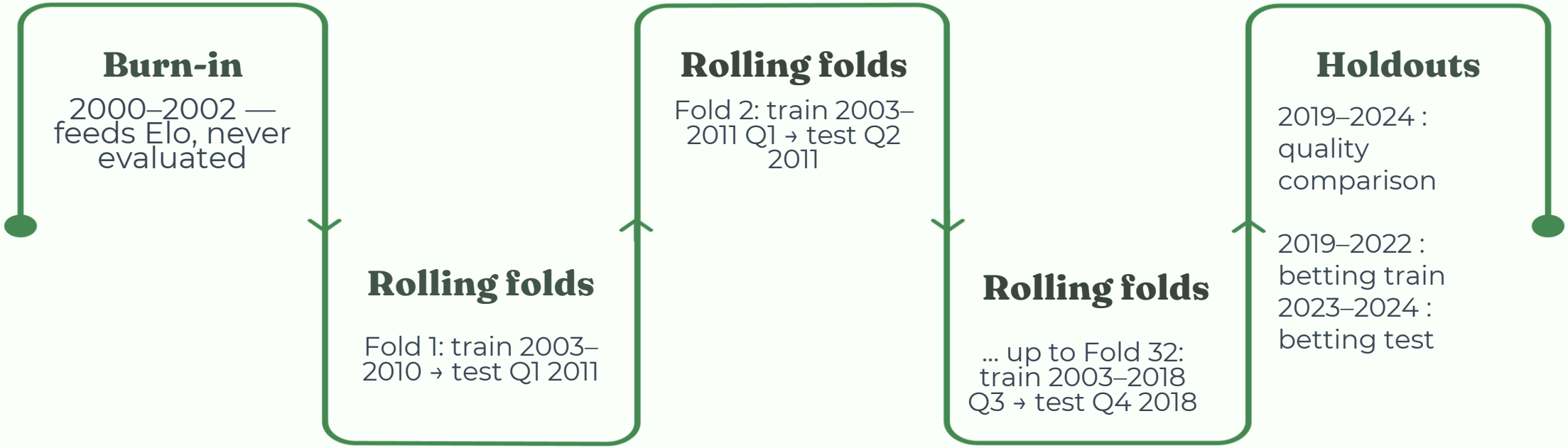
Two design choices

Two parallel systems. One global rating, one per surface, combined linearly with weight λ. A grid search on the training period alone selected λ = 0.5.

A K-factor that decays. K = k_0 / (n + offset)^shape, where n is matches played. Newcomers move fast, veterans stay stable.

Elo alone as a baseline: **68.92% accuracy | 0.5864 log-loss**

Rolling-origin cross-validation



The training window expands; the test quarter always lies in the future. 32 folds.

Why not k-fold

Shuffling matches would let the model train on 2018 to predict 2011. Tennis has a direction; the validation must respect it.

Why expanding, not sliding

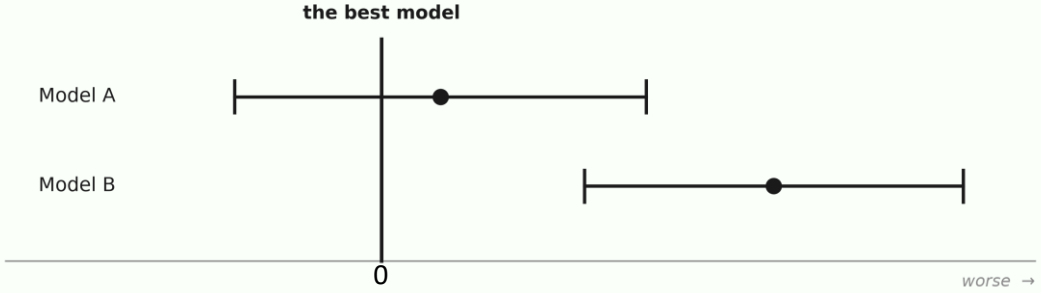
Elo and form features are cumulative. Discarding old matches would throw away the history the features are built from.

Why quarterly

A quarter is long enough to contain several tournaments and short enough to give 32 folds from eight years.

Model comparison

Each model is compared with the best one, fold by fold. The 32 differences in log-loss give one confidence interval.



Model A: the interval contains zero

In some folds A was ahead, in others behind. We cannot say which model is better. But we can say the difference is no larger than the far end of the bar and that bound is the result.

Model B: the interval excludes zero

B was worse in almost every fold. The difference is real, and it is at least the near end of the bar.

· Dot = mean of the 32 differences. · Bar = 95% confidence interval
· Caveat: folds share training data, so the bars are slightly narrow.

$$\bar{d} \pm t_{31,0.975} \frac{s_d}{\sqrt{32}}$$

Eleven classifiers

Seven Families

Linear

Logistic L1
Logistic L2

Probabilistic

Gaussian naive Bayes

Instance-based

k-NN (k=100)

Trees

Decision tree

Ensembles

Random forest
Gradient boosting
AdaBoost

Kernel

Linear SVM
RBF SVM

Neural

Feed-forward NN (64, 32)

Nothing was tuned. Every model runs on its standard scikit-learn configuration, fixed once and never searched over.
Tuning each to a different degree would confound algorithm quality with search effort the model that got the most attention would win, and the comparison would measure the author, not the algorithms.

Betting test

Accuracy measures predictions. The market measures whether they are worth anything the price already contains information the model cannot see.

$$EV = p \cdot o - 1$$

Bet when EV > τ

Expected profit per unit staked. p is the model's probability, o the decimal odds. We bet one side per match whichever has the higher expected value and only if it clears the threshold τ.

$$1/oA + 1/oB = 1.0262$$

Bookmaker's margin: 2.62%

Implied probabilities should sum to 1. They sum to 1.0262. The extra 2.62% is the bookmaker's margin on the test years, the bar the model must clear before any edge exists.

Why we use the raw odds

We could divide the margin out and recover what the bookmaker really believes. We do not, because the odds actually paid are the raw ones. Removing the margin would lower the bar, place more bets, and show returns nobody could collect.

So, matching the bookmaker is not enough.
The model has to beat him by more than 2.62%.

How the simulation is run

One unit per bet

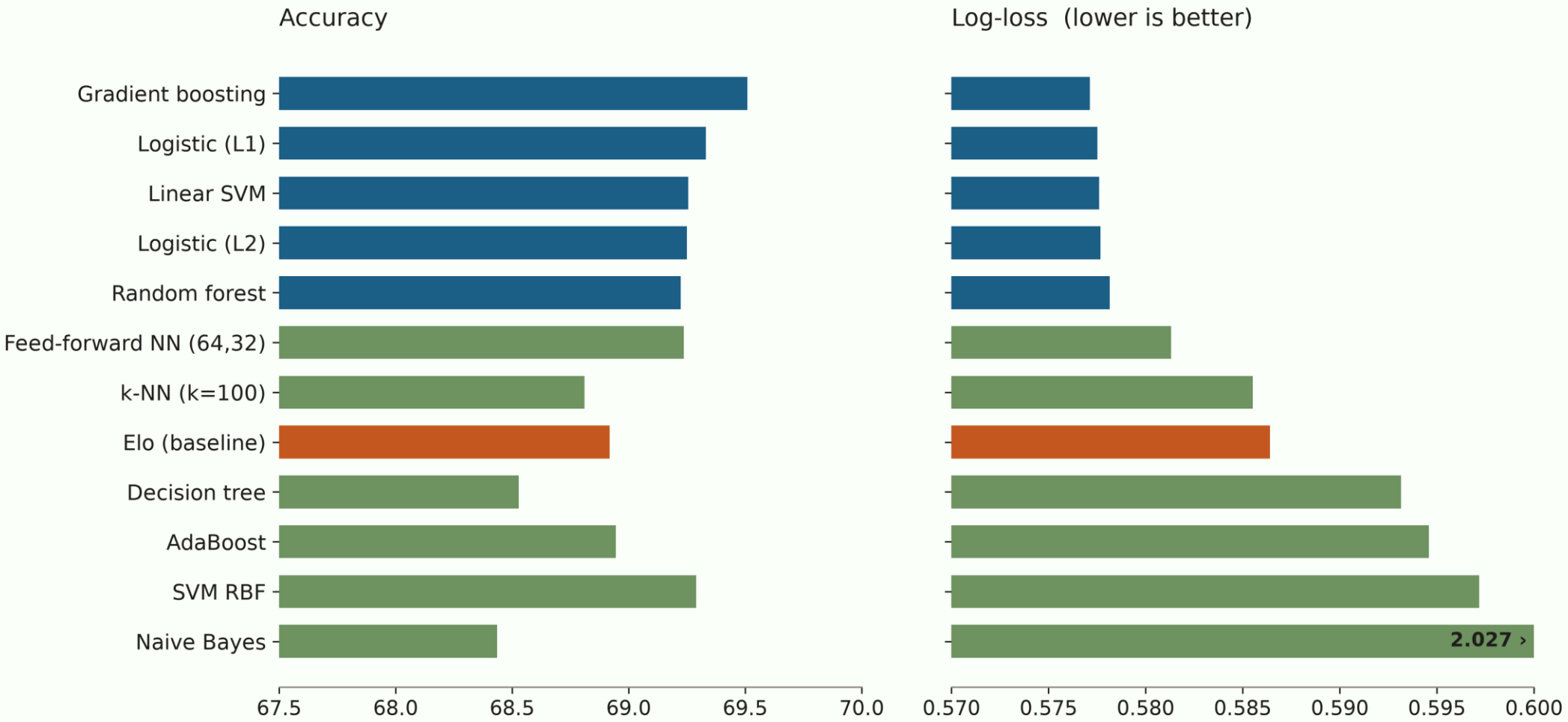
No stake sizing, so the return reflects what the model knows rather than how the bankroll was managed.

Seven thresholds, τ from 0 to 0.30

A high τ bets only where model and market disagree strongly.

Classifiers results

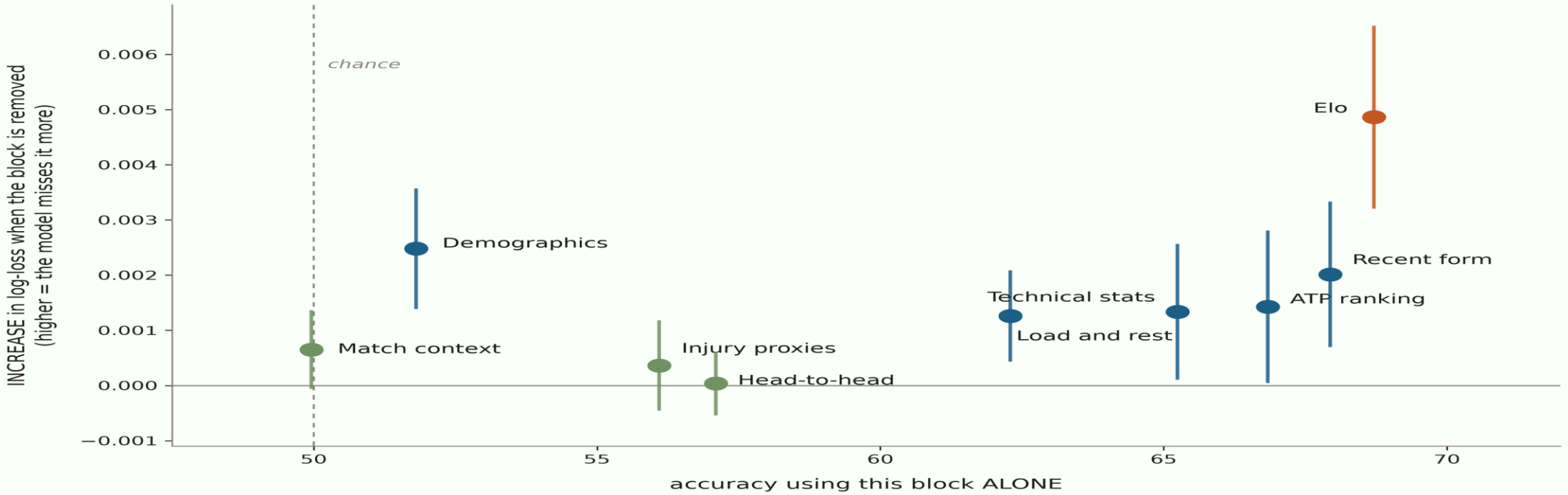
Eleven classifiers, 32 quarterly folds



Five models converge in the fourth decimal place. Elo is 0.0093 away, more than four times the most that separates them from each other (0.0021). The same analysis on accuracy agrees: seven models tied within 0.3 points, Elo beaten by 0.59. Everything that follows uses gradient boosting, first in the ranking.

Prediction

Nine feature blocks. Horizontal: accuracy using that block alone. Vertical: increase in log-loss when it is removed from the full model.



Demographics predicts nothing, matters anyway

51.8% alone, barely above chance. Yet removing age, height and handedness costs 0.0025. They do not predict; they modulate the other features.

Head-to-head contributes nothing

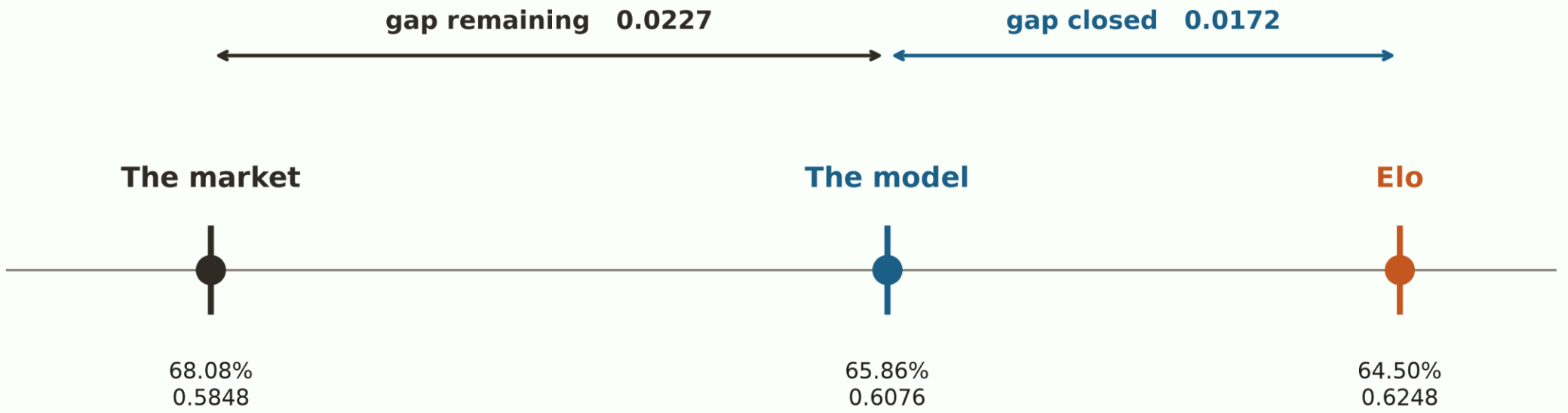
57.1% alone, but its interval is [−0.0005, +0.0006]. Elo and recent form already carry the information. Of 35,368 distinct pairs, 62.9% have met exactly once.

Match context

49.96% alone. Surface, round and format cannot say WHO wins.
Any departure from 50% would have signalled a defect in the symmetric representation.

Beat Elo, lose to the market

Trained on 2003-2018. Evaluated on everything after it: 2019–2024, the 12,824 matches for which market odds exist.
"The market" is the bookmaker's odds converted to a probability, with the margin divided out so the two sides sum to one.



0.0172 closed

1.4 accuracy points over Elo, and Elo was beaten in 29 of the 32 cross-validation folds.
This is what 38 features and a full model search bought.

0.0227 remaining

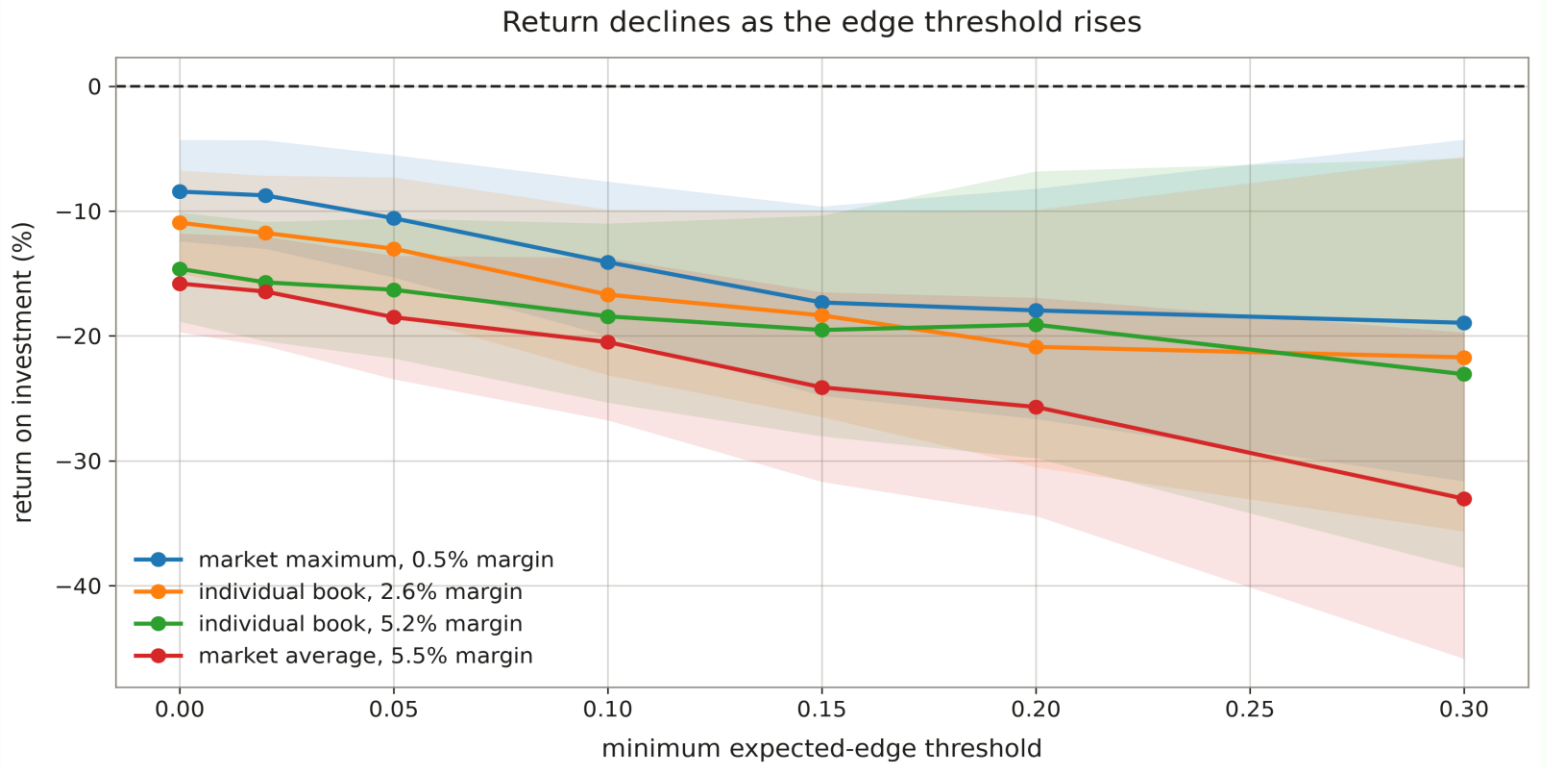
Larger than what was closed.
The bookmaker is further ahead of the model than the model is ahead of Elo.

Less than half the distance. And the ceiling is not the algorithm, eleven of them agreed.

The margin is divided out here because comparing probabilities requires a valid distribution. It is not divided out in the betting simulation, where the price actually paid is what matters.

Betting test results

2023–2024. One unit per bet, four odds sources, seven thresholds.



The margin is not the cause.

At the best price available anywhere the margin is 0.5%, and the loss is still 8.45%.

Every return is negative.

All four sources, all seven thresholds, and every bootstrap interval stays below zero. Not bad luck.

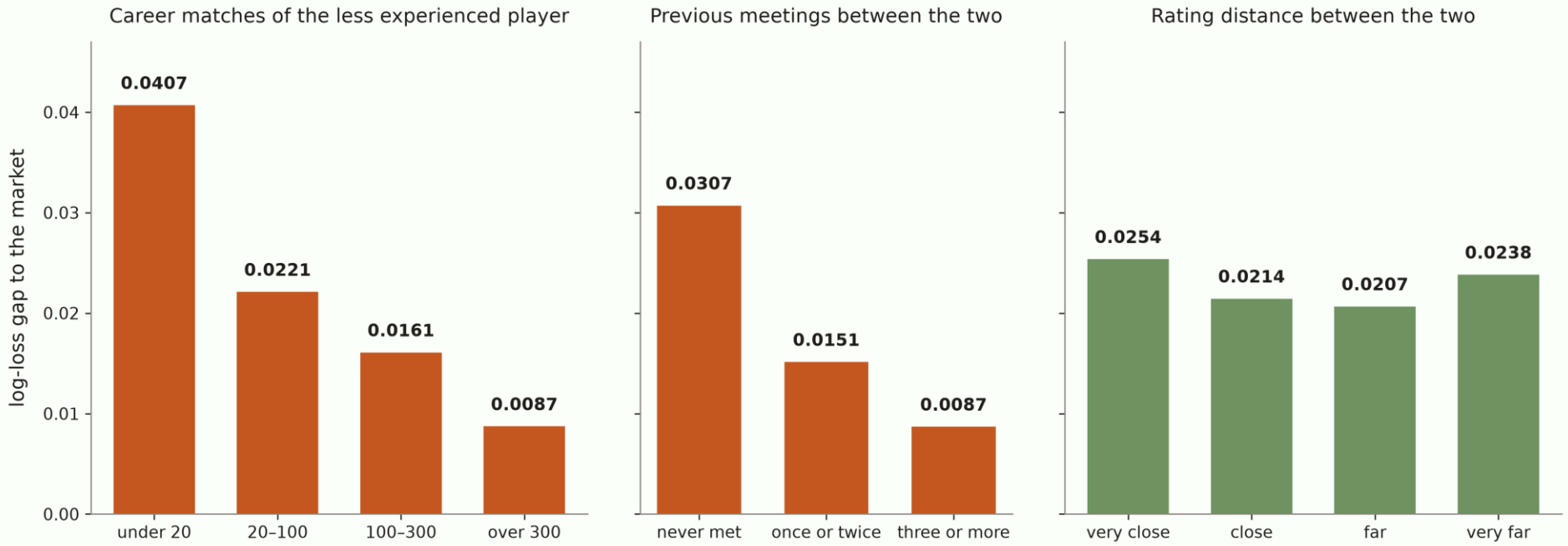
It gets worse, not better.

A higher threshold bets only where model and market disagree most. Real information would improve the return there. It falls instead.

The model bets mostly on underdogs and overvalues them.

The gap

The market wins in all 29 segments examined. The interesting part is by how much.



History predicts the gap

0.0087 when both players have long careers and have met before. 0.0407 when one of them has fewer than 20 career matches, almost five times larger.

Difficulty does not

Split the same matches by how evenly the two are rated and the gap barely moves: 0.0207 to 0.0254.

Close matches are harder for everyone, including the bookmaker.

This is an information deficit, not a modelling one.

Conclusion

The algorithm does not matter

Five classifiers of very different inductive bias converge to the same performance. Whatever separates them is at most 0.0021 of log-loss, under a quarter of what separates them from Elo.

The representation does

A symmetric, verifiably causal feature pipeline puts all five clearly above a tuned surface-blended Elo. Three negative results came with it: head-to-head adds nothing, injury proxies add nothing, and the market cannot be beaten.

The ceiling is the data

All five sit clearly below the betting market. The gap is 0.0087 where match history is dense and 0.0407 where it is thin and flat across difficulty.

The binding constraint is the information in result data.

The remedy lies in different sources: current physical condition, and information on players with thin history.

Limitations

The betting simulation is not a strategy.	Closing odds, not always obtainable at decision time. No stake limits, no commissions. The returns measure informational content relative to the market, and nothing more.
Nothing was tuned.	Deliberate, to keep the comparison between families fair. A tuned model would likely gain a little, though five model classes converging suggests very little.
The folds are not independent.	The training windows expand, so consecutive folds share almost all their data. The confidence intervals are therefore slightly narrower than they should be.

THANKS FOR YOUR ATTENTION

Antonio Cannito · **a.cannito21@studenti.uniba.it**